

Aditi Ravi Kumar

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EDUCATION

Columbia University

New York, USA

Master of Science in Biomedical Engineering, GPA: 3.92

2025–2026

Relevant Coursework: Embedded AI (Spring '26), Brain-Computer Interfaces Lab, Robotics Studio, ML for Genomics, Healthcare Operations Management

BMS College of Engineering

Bangalore, India

Bachelor of Engineering in Biotechnology, GPA: 9.52/10 (Gold Medalist)

2021–2025

RESEARCH EXPERIENCE

Research Assistant, Columbia Business School

2025–Present, NY, USA

- Analyzed CMS Medicare claims and outcomes data (1TB+) across 6M+ patient records to evaluate pharmaceutical impact on mortality, treatment effectiveness, and population-level health outcomes using Python, SQL, and regression modeling
- Applied ETL pipelines, data cleaning, and SQL-based queries on Redvis to extract Medicare reimbursement patterns and identify trends in treatment effectiveness and care quality metrics

Project Intern, Indian Institute of Science (IISc) - Saini Lab & Mathematical Oncology Lab

2024–2025, India

- Investigated mitochondrial dysfunction as a mechanism underlying senescence-induced EMT state transitions; integrated findings across computational and experimental approaches
- Designed and executed bioinformatics pipelines for bulk and single-cell RNA-Seq analysis across 15 GEO datasets, identifying conserved molecular signatures in EMT, hypoxia, and senescence pathways
- Built ODE-based signaling network models in Python to predict dynamic cellular responses; reduced data processing time by 60% through automated R pipelines
- Performed wet lab experiments: mammalian cell culture (A549, BEAS-2B), flow cytometry, immunofluorescence, protein extraction and quantification (Bradford, BCA), Western blotting, SDS-PAGE

Research Intern, Sri Shankara Cancer Foundation

2023, India

- Designed primers for 8 genes validating expression data for bladder cancer using PCR; achieved 70% validation success rate
- Analyzed clinical data from 71 breast cancer patients using R to stratify tumor subtypes and identify prognostic biomarkers
- Performed DNA/RNA extraction, quantification, and gel electrophoresis for molecular biology experiments

SELECT PROJECTS

NanoMaps: Nanobot Tracking System in Bloodstream | Columbia Healthcare Hackathon – Advanced Therapeutics Track Winner

2026

- Designed a real-time tracking and navigation system for nanobots within the bloodstream, targeting precision drug delivery applications
- Won the Advanced Therapeutics track at Columbia University's Healthcare Hackathon 2026

Quadruped Walking Robot | Robotics Studio, Columbia University

Fall 2025

- Built a quadruped walking robot from the ground up, including circuit design, PCB layout, and hardware assembly
- Implemented gait optimization algorithms and control systems for stable locomotion

Brain-Computer Interface Systems | BCI Lab, Columbia University

Fall 2025

- Developed signal processing pipelines for P300 and motor imagery classification, achieving 87.81% accuracy
- Implemented SSVEP analysis for brain-computer interface applications using EEG data

Integrative Multi-tissue Analysis of Rheumatoid Arthritis Biomarkers

2024

- Integrated transcriptomic datasets from whole blood and synovial tissue; performed pathway enrichment, single-cell deconvolution, and epitope analysis to predict immune cell-associated biomarkers

DBSCAN Clustering for Glioblastoma Metabolomic Data

2024

- Implemented and optimized DBSCAN clustering on glioblastoma metabolomic data to identify treatment-naïve-specific metabolite patterns (GitHub)

LRRK2 Drug Discovery for Parkinson's Disease

2023–2024

- Applied computational screening and molecular dynamics simulations to identify promising drug candidates

Batch-Aware AMICI for Spatial Transcriptomics

2024

- Implemented batch effect correction methods for AMICI spatial transcriptomics analysis framework to improve data integration (GitHub)

TECHNICAL SKILLS

Programming & Data Analysis: Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), R (Tidyverse, DESeq2, Seurat, edgeR, limma, Bioconductor, clusterProfiler, biomaRt, ggplot2, dplyr), SQL, MATLAB, Linux, Git/GitHub

Hardware & Design: AutoCAD (3D modeling), 3D Printing, Prototyping, EEG Signal Acquisition & Processing, Raspberry Pi, Embedded Systems Programming

Bioinformatics & Computational Biology: Bulk RNA-Seq, Single-cell RNA-Seq (scRNA-seq), Spatial transcriptomics, Differential expression analysis (DESeq2, edgeR, limma), Pathway enrichment (GSEA, clusterProfiler), GEO datasets, Meta-analyses, Transcriptomic data integration, Molecular dynamics (GROMACS), PyMOL, Modeller

Machine Learning & Statistics: Statistical Modeling, Regression Analysis, Survival Analysis, Clustering (DBSCAN), Supervised/Unsupervised Learning, ODE-based Simulations

Wet Lab & Biochemistry: Mammalian cell culture (A549, BEAS-2B), Flow cytometry, Immunofluorescence, Protein extraction & quantification (Bradford, BCA), Western blotting, SDS-PAGE, DNA/RNA extraction, PCR, Primer design, Gel electrophoresis

LEADERSHIP

Mentors & Experts Lead , Nucleate New York	2025–Present
Evaluate biotech accelerator applications and startup opportunities; coordinate entrepreneurship events, connect experts in various biotech fields to startups	
Vice President , Synapse Biotechnology Association	2024–2025
Led cross-functional teams organizing workshops and events for 200+ students; managed logistics, budgets, and communications	

PUBLICATIONS

First Author: Molecular Dynamics Studies on Herbal Contraceptives as Progesterone Receptor Modulators. *ICIIBMS 2024, IEEE Xplore*

Co-Author: Non-coding RNA in Head and Neck Squamous Cell Carcinoma: A Comprehensive Review. *Under Review, Heliyon*

Lead Author (Posters): Integrative Multi-tissue Analysis of Rheumatoid Arthritis Biomarkers; Cell & Gene Therapy Landscape in India. *27th Bengaluru Tech Summit, 2024*

AWARDS

Healthcare Hackathon Winner , Advanced Therapeutics Track, Columbia University	2026
Certificate of Merit (8x), Quarter tuition waived per award	2022–2025
Best Paper Award , ICIIBMS Conference, Okinawa, Japan	2024
Training: Neural Networks & Deep Learning (Deeplearning.ai) Cell Therapy Manufacturing (MITx)	